

PRAJWAL N

prajwalnvs1000@gmail.com – +91 90199 15516 –

photo.jpg

SUMMARY

Final-year B.E. Electronics and Instrumentation Engineering student at JSS Science and Technology University with hands-on experience in Embedded Systems, Digital Signal Processing, IoT, and Industrial Automation. Skilled in Python, C, and C++ with practical experience in ESP32, Arduino, sensors, microcontrollers, PLC systems, and signal processing. Interested in Embedded Systems, Electronics, Instrumentation, IoT, and Automation.

EDUCATION

JSS Science and Technology University, Mysore B.E. Electronics and Instrumentation Engineering	2023–2027
GSI PU College, Mysore PCMCs	2021–2023 Percentage: 84.8%
GSI School, Mysore SSLC	2020–2021 Percentage: 69.9%

PROJECTS

AI-Based Drone Detection and Direction Estimation System • Developed a portable embedded system for real-time drone detection using acoustic signal processing. • Implemented band-pass filtering, FFT, MFCC feature extraction, and cross-correlation for sound classification and direction estimation. • Integrated OLED display, buzzer, and Wi-Fi alerts for real-time surveillance applications.	<i>ESP32-S3, INMP441, DSP</i>
Adaptive Eating Assistance System • Developed an embedded assistive robotic feeding system for elderly and physically challenged individuals. • Integrated ESP32, MPU6050, IR eye-blink sensor, and servo motors for real-time sensor acquisition and motion control. • Designed the system to support hands-free feeding through automated robotic arm movement.	<i>ESP32, MPU6050, IR Sensor, Servo Motors</i>
Vibration and Tilt Measurement using MPU6050 • Developed a real-time vibration and tilt monitoring system using MPU6050 accelerometer and gyroscope. • Processed sensor signals to estimate orientation and monitor vibration for embedded applications.	<i>ESP32, Embedded Systems</i>
PLC-Based Predictive Maintenance System • Designed a predictive maintenance system for industrial motors by monitoring vibration and temperature. • Implemented fault detection, alarm generation, and HMI monitoring to improve equipment reliability and reduce downtime.	<i>PLC, Industrial Automation</i>

TECHNICAL SKILLS

Programming: Python, C, C++, SQL
Embedded Systems: ESP32, Arduino, Microcontrollers, Sensor Interfacing, IoT, Servo Control
Digital Signal Processing: FFT, MFCC, Digital Filtering, Signal Analysis, Acoustic Signal Processing
Industrial Automation: PLC, HMI, PID, Predictive Maintenance, Fault Detection
AI/ML: Machine Learning, Scikit-learn, Pandas, NumPy, Data Preprocessing
Web Technologies: HTML, CSS, JavaScript, React.js, Node.js, Express.js, Flask
Databases: MongoDB, SQLite
Tools: Git, GitHub, VS Code, MATLAB, Chart.js

ACHIEVEMENTS

- NCC 'A' Certificate holder and recognized as Best Cadet.
- Participated in Republic Day Camp and Level-4 NCC Camps.
- Earned Senior Level Tabla Certification and performed at multiple cultural events.

CLUB ACTIVITIES

- Active member of the Hardware Club, JSS Science and Technology University; participated in embedded systems workshops and electronics hardware development activities.
- Active member of the Naada Club; performed Tabla at university cultural events.

LANGUAGES

English, Kannada, Hindi, Telugu